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Population genetics of walleye and yellow perch in the St. Marys River

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ABSTRACT

We assessed the genetic diversity of walleye (*Sander vitreus*) and yellow perch (*Perca flavescens*) in the St. Marys River system. Walleye samples were collected from four current spawning populations, one historic population, and the two sources used for stocking the St. Marys River. Non-spawning perch were collected from ten sites along the St. Marys River. Five microsatellite loci were used for screening each species to create a genetic baseline to which future populations can be compared and to assess the impacts of past stocking on the walleye populations. Resulting allele and genotype frequencies were characterized to quantify genetic variation and differentiation of the respective populations. Both the walleye and yellow perch populations in the SMR exhibited some degree of population structure. Although stocking has modified the genetic diversity of walleye, there is still at least one unique population (Bar River, Ontario) remaining in the St. Marys River system. Both species had unique populations in Lake George, suggesting that this area may have been isolated in the recent past and should be protected from future stocking events to preserve this uniqueness. Further work is encouraged to understand the nature of this possible genetic isolation.

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Introduction

The St. Marys River (SMR), the sole outlet of Lake Superior, forms the connecting waterway to Lake Huron, and contains the international boundary between Michigan and Ontario. The river boasts some of the clearest water in the world, being fed by Lake Superior, the most oligotrophic of the Great Lakes (Trebitz et al., 2007). In 1985, the SMR was identified as an area of concern by the US Environmental Protection Agency, with nine beneficial use impairments. One of these impairments, the degradation of fish populations, is addressed in this paper using genetic information to assess the diversity of the river's walleye (*Sander vitreus*) and yellow perch (*Perca flavescens*) stocks.

Walleye and yellow perch are native to the SMR and serve important roles both ecologically and economically as targets of substantial fishing effort. Depending on their life-history stage, walleye and yellow perch feed at a variety of different trophic levels. Young-of-the-year of both species mainly feed on zooplankton (Luecke et al., 1990; Mills et al., 1989), but later switch to feeding on benthic macroinvertebrates, crustaceans, and small fish (Keough et al., 1996). Walleye become piscivorous at an earlier age than yellow perch, and also utilize the perch as a forage species (Hartman and Margraf, 1993). Yellow perch support a popular recreational fishery, and also are harvested commercially in some areas of the Great Lakes (Francis et al., 1996). Walleye are one of the most popular game fish species in the SMR, and fisheries managers have supplemented native populations for over 25 years with stocking. Understanding the status of these two species is therefore important for both anglers and the managers who seek to provide healthy, stable populations for public use.

Populations of yellow perch have been relatively stable in the SMR since the 1970s (Schaeffer et al., 2011) and walleye populations have been supplemented with stocking; however, little is known about the genetic diversity of these species in this river. Genetic information has proven viable and useful for understanding the population dynamics within changing environments (Garant et al., 2000). Over time, fish populations become genetically adapted to their particular environment, and genetic diversity can be used as an indicator of a population's genetic history. A degraded ecosystem can reduce the effective population size, which will reduce the genetic diversity of the population (Luikart et al., 1998). As population size is reduced, allele frequencies will drift due to a smaller number of breeders, eventually leading to fewer alleles in the population and a decrease in heterozygosity (Cornuet and Luikart, 1996). Microsatellite DNA loci have been used extensively to assess genetic structure of populations (Miller, 2003; Strange and Stepien, 2007; Stepien et al., 2009), and they were chosen as the markers to assess the genetic status and to establish a baseline for future monitoring of the yellow perch and walleye populations in the SMR.

The genetic composition of fish populations can be influenced significantly by stocking. Although yellow perch, to our knowledge, have not been stocked into the SMR, regular walleye stocking has occurred since 1985 (Godby et al., 2009). At that time, hatchery

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personnel were not concerned with genetically matching the source and recipient populations. In these situations, alleles introduced into a population can swamp local alleles, causing loss of native genetic diversity (Miller and Kapuscinski, 2003). Walleye have been stocked into the SMR from two different sources outside the SMR system, Bay de Noc, Delta County, Michigan and Manistique Lake, Mackinac County, Michigan. Fish from Bay de Noc were stocked on a larger scale and for a longer period of time than fish from Manistique Lake. Since 1999, walleye originating from Munuscong Bay, within the SMR, have been used to stock the system.

It is difficult to assess the origins of walleye in the SMR, because stocked fish have not been consistently marked. Most fish stocked in the 1980s and 1990s were not marked at all; however, more recent stockings have included batch marking with oxytetracycline (OTC). The success of these stockings has been assessed on a limited basis by examining OTC marks on juvenile walleye otoliths collected by the Chippewa Ottawa Resource Authority (CORA). These data suggested that stocked fish do contribute to the juvenile walleye populations in the SMR (M. Ebener, CORA, personal communication). Our goal was to measure the influence of stocking on adult spawning fish, in order to determine whether stocking has changed the genetic signature of the walleye from that of the native population (Caroffino, 2004). Because gene flow does not occur naturally between the two source populations used for stocking and that in the St. Marys River, assuming sufficient differentiation among the respective populations, then genetic analysis of the current spawning populations should allow determination of any influence from the source populations. The objectives of this study were to establish a genetic baseline for walleye and yellow perch, and to determine the impacts of stocking on the genetic diversity of walleye in the SMR.

Methods

Sample collections

Walleye were collected from four spawning populations throughout the SMR during 2002 and 2003 (Fig. 1). The Potogannissing River is located in the southern portion of the SMR and contains a spawning population that, according to local residents, surged after stocking in the 1980s. The Bar River is a Canadian tributary of the SMR that supports a native spawning population and flows into Lake George, a lacustrine portion of the St. Marys River. Munuscong Bay is a large, shallow bay in the central portion of the SMR and contains the largest lentic-spawning population of walleye in the SMR. The Waiska River is

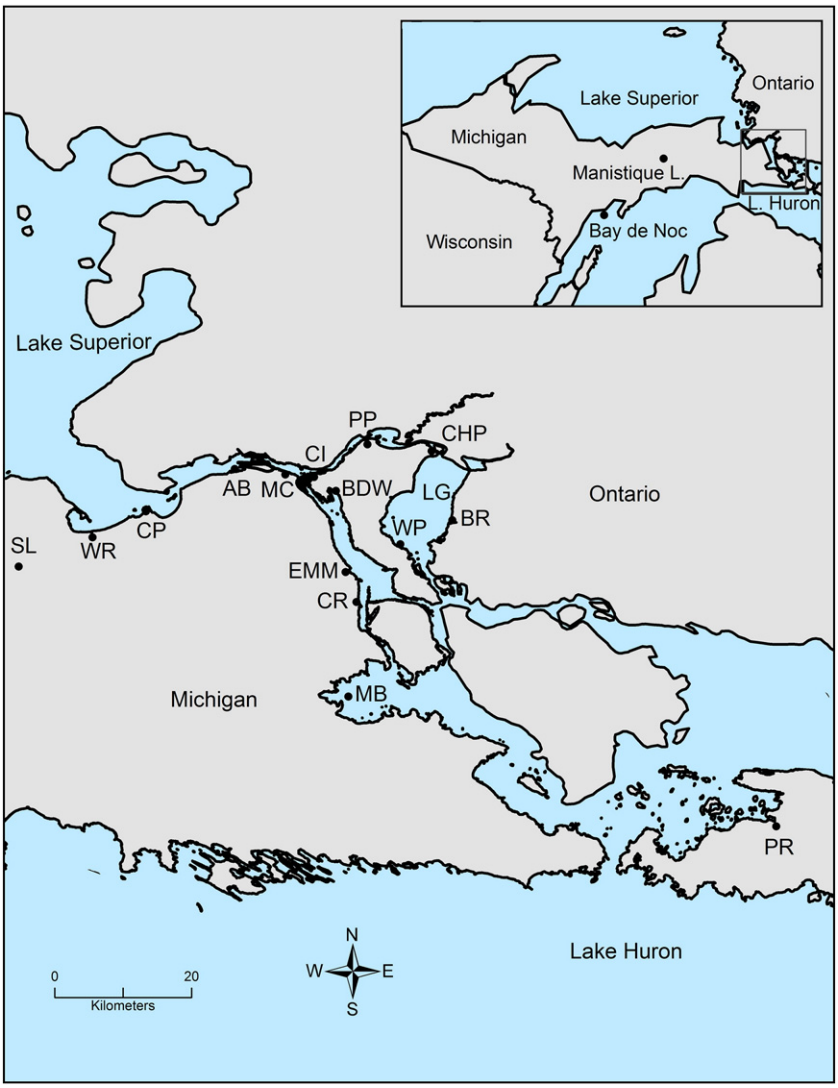

Fig. 1. Map of the St. Marys River showing locations of sites sampled and Lake George (LG). Walleye sampling locations underlined: Bar River (BR), Charlotte River (CR), Munuscong Bay (MB), Potogannissing River (PR), and Waiska River (WR). Inset map shows locations of origins for stocked walleye from Manistique Lake and Bay de Noc. Yellow perch sampling locations in italics (*YP*): 11 Mile Marsh (*EMM*), Palmer's Point (*PP*), Ashmun Bay (*AB*), Soldier Lake (*SL*), Bai de Wasai (*BDW*), Churchville Point (*CHP*), Whipple Point (*WP*), Cook Island (*CI*), Cedar Point (*CP*), and Mission Creek (*MC*).

located in the upper SMR and supports the largest spawning population of walleye above the Sault Locks. Mature walleye were collected from each of these populations during the spawning season, using dip nets and fyke nets. All captured fish had a portion of a fin ray removed, dried, and stored at room temperature for later genetic analysis.

In addition to the contemporary walleye spawning population samples collected during 2002 and 2003, historic scale samples were obtained from the Charlotte River, a tributary located near the SMR shipping channel. These samples were collected during spawning runs in the early 1980s and were archived at the Lake Superior State University Aquatic Research Laboratory. During 2002 and 2003, the spawning population in the Charlotte River was small, and we were unable to obtain any additional samples.

For comparison to the contemporary SMR spawning populations, tissue samples also were collected from the two walleye source populations used to stock the SMR. Natural resource agency personnel provided fin samples from walleye in spawning condition from Bay de Noc, a northern Lake Michigan population, and scale samples were obtained from spawning walleye in Manistique Lake, an inland lake located in the central Upper Peninsula of Michigan.

Yellow perch samples were collected as part of a larger project assessing the biotic integrity of the SMR through analysis of the river's coastal marshes. Yellow perch were captured during the summer of 2005 from nine sites along the SMR (Fig. 1). Sample sites were located in the upper SMR (Cedar Point and Ashmun Bay), the lower St. Marys shipping channel (Mission Creek, Bai de Wasai, and 11 Mile Marsh), and the north channel leading to Lake George (Cook Island, Palmers Point, Churchville Point, and Whipple Point). Yellow perch samples also were collected from Soldier Lake to be used as an outgroup. Soldier Lake is an inland lake isolated from the SMR, located about 50 km west of Sault Sainte Marie, Michigan. Samples were collected using fyke nets at each site in either June or July 2005. From each captured fish, a fin ray was removed, dried, and stored at room temperature for later analysis.

Laboratory methods

DNA from both species was extracted using a 10% Chelex (Carolina Biological, Burlington, North Carolina)/0.01% Proteinase K (Invitrogen, Carlsbad, California) solution. Approximately 10 mg of dried fin membrane was placed in 300 μL of the extraction solution and was incubated in a 37 °C water bath for 10 min. Each sample then was vortexed before being placed into a boiling water bath for 8 min. After cooling, samples were spun for 2 min in a microcentrifuge, and the supernatant (200 μL) was removed and stored at −20 °C.

Extracted DNA samples were prepared for gene amplification using the following polymerase chain reaction (PCR) reagents: 0.3 μM fluorescently labeled primer (Applied Biosystems, Inc., Foster City, California), 1.5 μL DNA template (concentration not determined), 10× PCR buffer (containing 15 mM $MgCl_2$, USB Corporation, Cleveland, Ohio), 10 mM nucleotide mix (USB Corporation), 0.5 U *Taq* polymerase (USB Corporation), and purified water up to a final volume of 50 μL. We used five microsatellite loci developed for yellow perch by Leclerc et al. (2000; Pfla L1, L2, L3, L5, L6), and five loci developed for walleye by Borer et al. (1999; Svi 4, 33, 17, 18) and Wirth et al. (1999; Svi L8; Table 1). Gene Amp® PCR System 9700 Thermocycler (Applied Biosystems) conditions for both species followed those of Borer et al. (1999) with primer-specific annealing temperatures (Table 1).

Amplified PCR products underwent capillary electrophoresis on an ABI Prism™ 310 Genetic Analyzer. The samples were prepared by combining 12.5 μL formamide, 0.5 μL Tamra internal size standard (Applied Biosystems, Inc. Foster City, California) and 1.0 μL of PCR product for a final volume of 14.0 μL. Samples were denatured at 95 °C for 3 min and then snap-cooled in an ice bath for 2 min before being loaded into the genetic analyzer.

Table 1
Characteristics of the microsatellite loci used in this study for walleye (Svi) and yellow perch (Pfla), including: number of alleles, the allele size range in base pairs, the observed (H_o) and expected heterozygosities (H_E), PCR annealing temperatures (T), and reference for each locus.

Locus	Alleles	Size range	H_o	H_E	T (°C)	Reference
Svi 4	11	102–126	0.76	0.72	60	Borer et al. (1999)
Svi 17	7	102–120	0.66	0.64	60	Borer et al. (1999)
Svi18	6	106–126	0.55	0.57	50	Borer et al. (1999)
Svi 33	12	80–106	0.77	0.79	50	Borer et al. (1999)
Svi L8	16	104–144	0.77	0.85	50	Wirth et al. (1999)
Pfla L1	21	98–134	0.59	0.70	50	Leclerc et al. (2000)
Pfla L2	14	198–232	0.50	0.57	53	Leclerc et al. (2000)
Pfla L3	22	134–178	0.58	0.75	53	Leclerc et al. (2000)
Pfla L5	7	130–154	0.33	0.36	53	Leclerc et al. (2000)
Pfla L6	19	134–178	0.56	0.57	52	Leclerc et al. (2000)

Data analysis

Alleles were scored using the software Genotyper (version 2.2, Applied Biosystems, Inc., Foster City, California), and MICRO-CHECKER (van Oosterhout et al., 2004) was used to determine whether scoring errors or null alleles were likely to be present. Population differentiation was assessed using pairwise Fst values calculated using the software Genepop (Raymond and Rousset, 1995). The program STRUCTURE (Pritchard et al., 2000) was used to infer the number of genetic groups (K) in both the yellow perch and walleye data sets. We tested values of K from one to seven for walleye and one to nine for yellow perch, and selected the most likely K based on the methods of Evanno et al. (2005). Once the appropriate K was selected, genotypes of individual fish were allocated to one or more genetic groups, without prior knowledge of their true group membership. Phylogenetic trees were constructed for walleye and yellow perch populations based on the genetic chord distance of Cavalli-Sforza and Edwards (1967). The neighbor component of the PHYLIP software package (Felsenstein, 2002) was used to construct each neighbor-joining tree, and their support was evaluated based on 1000 bootstrap resamplings using the Seqboot component.

Results

Walleye

At least 32 fish were sampled from the spawning populations in the SMR, as well as the stocked source populations. However, only 19 samples were available from the Charlotte River. Scoring errors or null alleles were not apparent at any of the loci examined. Polymorphism was observed at all loci, and heterozygosities were reasonably high (Table 2). Bar River and Charlotte River had the smallest numbers of alleles (24 and 28, respectively) across the five loci; all other populations had in excess of 33 alleles (Table 2).

Even though nearly all Fst values among walleye populations were significant, the magnitude of the Fst values differed and some patterns emerged (Table 3). The walleye population spawning in the Bar River

Table 2
Genetic variation within the seven walleye populations sampled for this study, including: number of samples from each population (N), aggregate number of alleles at the five loci (A), and observed (H_o) and expected (H_E) heterozygosities across all five loci.

Population	N	A	H_o	H_E
Potogannissing River (PR)	33	39	0.68	0.72
Bar River (BR)	35	24	0.57	0.56
Munuscong Bay (MB)	34	37	0.63	0.69
Waiska River (WR)	33	35	0.70	0.70
Bay de Noc (BDN)	36	34	0.74	0.70
Manistique Lake (ML)	32	33	0.58	0.58
Charlotte River (CR)	19	28	0.53	0.53

Table 3
Pairwise Fst values for walleye populations: Bay de Noc (BDN), Bar River (BR), Charlotte River (CR), Manistique Lake (ML), Munuscong Bay (MB), Potogannissing River (PR), and Waiska River (WR).

Population	BDN	BR	CR	ML	MB	PR
BR	0.1009[a]					
CR	0.1052[a]	0.0629[a]				
ML	0.0311[a]	0.0886[a]	0.1018[a]			
MB	0.0410[a]	0.0399[a]	0.0469[a]	0.0327[a]		
PR	0.0167[a]	0.0868[a]	0.0728[a]	0.0245[a]	0.0157[a]	
WR	0.0273[a]	0.0676[a]	0.0661[a]	0.0354[a]	0.0051	0.0014

[a] Significant following sequential Bonferroni correction for multiple tests.

showed a high degree of differentiation from both source populations (Fst > 0.089), whereas other spawning populations showed a lower degree. When compared to the source populations, Potogannissing River, Waiska River, and Munuscong Bay all had low Fst values ranging from 0.016 to 0.041. That is, these current populations more closely resembled the sources of stocking than they did the native population in the Charlotte River. Other than that in Bar River, there was little differentiation between the spawning groups within the SMR (Table 3).

The results of analysis using STRUCTURE estimated that the complete set of walleye genotype data was composed of three distinct genetic groups (K = 3; Fig. 2a). These results also indicated that Bar River was a unique population, as most individuals had the predominant signature of a single genetic group. Individuals from Charlotte River were mostly mixed between the Bar River signature and one that was common throughout the SMR populations. Munuscong Bay, Potogannissing River, and Waiska River resembled the source populations Bay de Noc and Manistique Lake, whereas Bar River and Charlotte River individuals did not (Fig. 2a).

The neighbor-joining tree diagram based on the genetic chord distances between these populations showed trends similar to those of the Fst and STRUCTURE analyses (Fig. 3a). The Bar River and Charlotte River populations were paired 89% of the time; Munuscong Bay was closest to these and paired with them 83% of the time. The other walleye populations did not group in the same way more than 45% of the time, indicating little support for genetic differences or structure among these populations.

Yellow perch

Yellow perch loci exhibited less polymorphism than was observed for walleye. Sample sizes for yellow perch ranged from 7 (Cook Island) to 61 (Palmer's Point) per population. The number of alleles across the five microsatellite loci varied between 18 (Ashmun Bay) and 40 (Palmer's Point), and mean heterozygosities across populations ranged from 0.36 (Bai de Wasai) to 0.64 (Soldier Lake; Table 4). Based on analysis using the MICRO-CHECKER program, null alleles were likely present at the Pfla L5 locus, and its data were removed from further analysis.

Two groups of yellow perch were distinct from all other samples. The perch collections from Churchville Point and Whipple Point, located in Lake George, were not differentiated from each other, but were significantly different from all other samples (Fst ranged from 0.077 to 0.225), even from that of the adjacent Palmer's Point. Although the Palmer's Point and Churchville Point sites are in close proximity to each other (Fig. 1), they were genetically distinct (Fst = 0.123, Table 5). There was little differentiation between the other perch samples in the SMR. As expected, the Soldier Lake population was significantly different from all SMR samples (Fst all > 0.191; Table 5).

From the neighbor-joining tree diagram, the yellow perch from Whipple Point and Churchville Point were paired 74% of the time, while perch from Soldier Lake were distinct in 93% of the resampling events (Fig. 3b). No other relationship was supported by more than 60% of the resampling events. The results of our analysis using STRUCTURE estimated that the complete set of yellow perch data was composed of four distinct genetic groups (K = 4, Fig. 2b). The Soldier Lake population was clearly distinct from all SMR populations. Churchville Point and Whipple Point were similar to each other, but distinct in genetic composition from all other sampled groups. Palmer's Point appeared to be a mixed population, with some individuals showing the signature of nearby Churchville and Whipple Points and others being more similar to fish sampled at Mission Creek or Bai de Wasai.

Discussion

We used genetic analyses to gain insights into the structure and history of walleye and yellow perch populations in the SMR. Both the

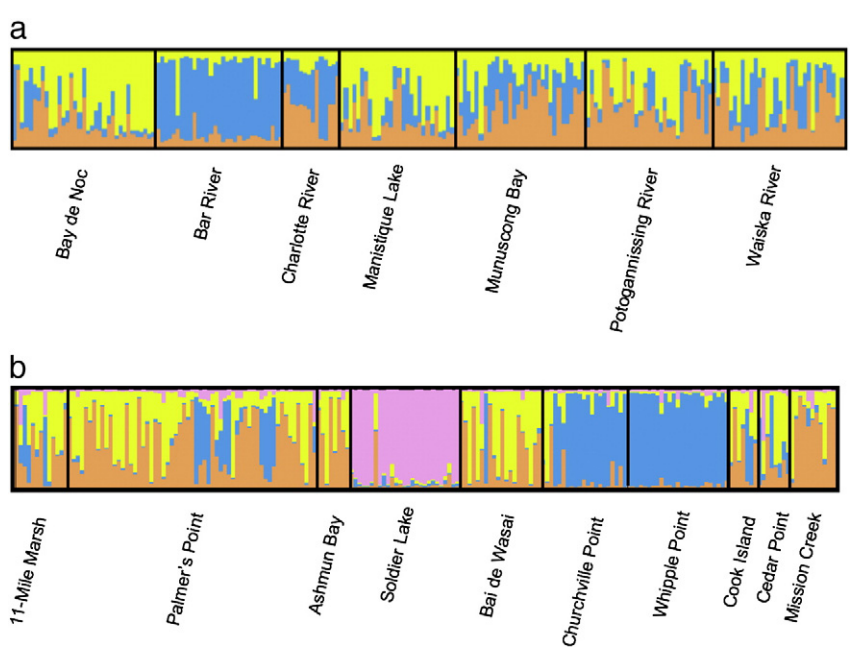

Fig. 2. Genetic composition of individual walleye (a) and yellow perch (b) from the sampled populations generated from the program STRUCTURE; shading represents the proportional membership of each individual to the three inferred genetic groups (K = 3 for walleye, and K = 4 for yellow perch).

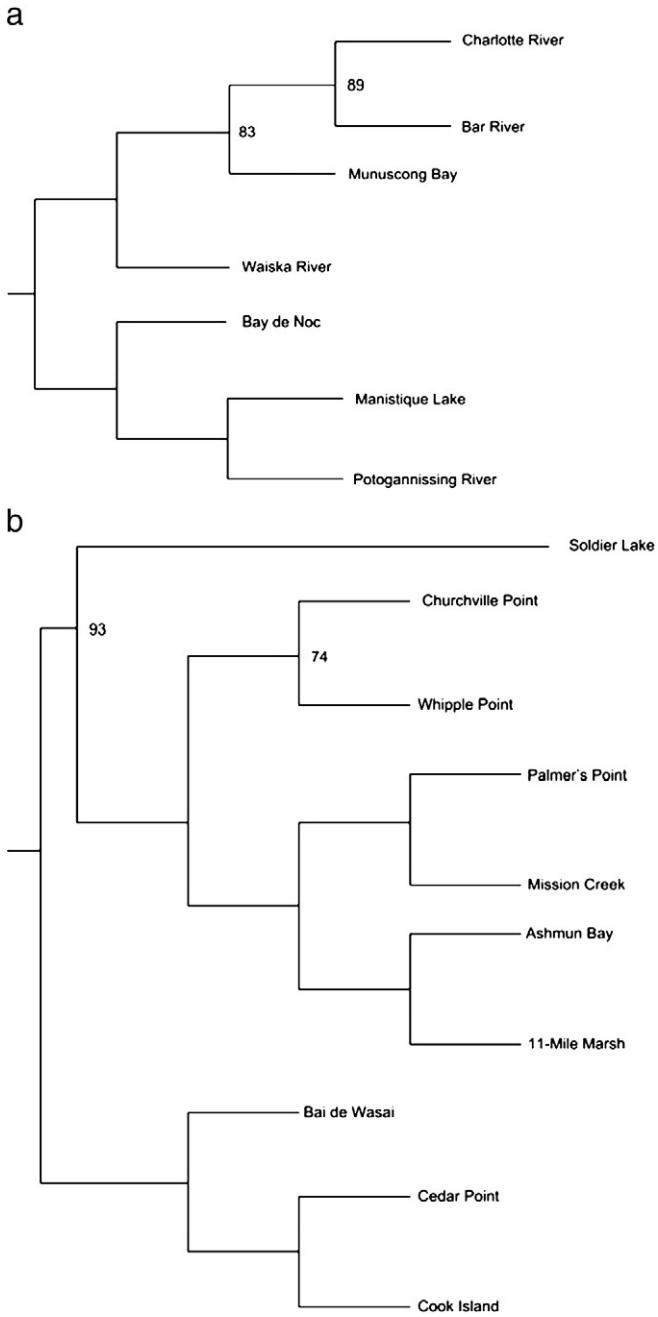

Fig. 3. Neighbor-joining tree diagram of relationships between walleye populations (a) and yellow perch populations (b) based on the Cavalli-Sforza and Edwards (1967) chord distance measure; the values at the nodes represent those relationships that had greater than 70% bootstrap support based on 1000 resamplings.

number of samples collected from each site (Tables 2 and 4) and the number of loci used in this study (5 per species) were lower than those used in many studies published in the current literature, and as a result, the conclusions drawn should be interpreted cautiously. However, these results have utility as the first to examine the genetic composition of walleye and yellow perch occupying the diverse system that is the SMR.

Walleye

Walleye populations in the SMR expressed a relatively large number of alleles at each microsatellite locus and heterozygosities that were similar to those documented in other populations throughout the midwest by Stepien et al. (2009), and in the southeast by Palmer et al. (2006). There is no historical genetic baseline to which the current populations in the SMR can be compared, nor historic samples that could be used to generate such a baseline. These data may serve as a first baseline, so that any changes can be assessed through comparison to future populations.

Despite significant Fst values between the source and spawning populations, stocking apparently has influenced certain walleye populations in the SMR. STRUCTURE results indicate little distinction between the sources of stocking and the spawning populations of SMR walleye in Munuscong Bay, Potogannissing River, and Waiska River. Significant Fst values may have arisen as a result of small sample sizes or from a founder effect when broodstock were selected for spawning. If either the broodstock that were selected or our sample of the source population did not adequately represent its variation, then genetic differences may be present between our sample of walleye from the source populations and the fish that were stocked (Miller and Kapuscinski, 2003). In addition, the type of stocking in the SMR may have led to the small differences observed between our samples of the source populations and those in the SMR. If fish were stocked into an area void of walleye, we could expect current samples to resemble the stocking source. However, walleye were stocked in a supplementation program, i.e., on top of existing populations. The stocked fish may have shifted the genetic signature of the SMR walleye, but not to the extent of giving rise to non-significant Fst values relative to the source populations. This outcome may be similar to that in Nipigon Bay in Lake Superior, where a walleye population at low abundance was supplemented by stocking, which resulted in a population that was genetically a mixture of stocked and presumably native ancestry (Wilson et al., 2007).

Although populations in some areas of the SMR have been influenced by stocking, walleye in the Bar and Charlotte Rivers did not resemble those in any other sampled population. Both of these rivers are used by walleye for spawning, but during other times of the year, walleye presumably mix in the SMR. These two populations both had fewer alleles than all other populations sampled, and although distinct from each other, they may reflect the historic range of genetic diversity in SMR walleye. The Bar River population, which does not resemble either stocking source, is likely a unique, native population that is persisting. This river is not geographically isolated from other populations in the SMR; however, fish have not been stocked in this location and Munuscong Bay is the closest stocking site. The observation of the unique Bar River population could parallel the situation in Cattaraugus Creek, a tributary to Lake Erie, where Stepien et al. (2004) found that a unique walleye population persisted despite supplementation with a different strain. It is unclear why the diversity of some unique populations persists in the face of stocking while that of others breaks down. Potentially their site fidelity (Stepien and Faber, 1998) has allowed them to adapt to the unique conditions in their natal rivers, yielding higher reproductive success than stocked fish. Palmer et al.

Table 4
Genetic variation within the ten yellow perch populations sampled for this study, including: number of samples collected from each population (N), aggregate number of alleles at the five loci (A), and observed (H_o) and expected (H_E) heterozygosities across all five loci.

Population	N	A	H_o	H_E
11 Mile Marsh (EMM)	14	27	0.52	0.65
Palmer's Point (PP)	61	40	0.47	0.57
Ashmun Bay (AB)	10	18	0.44	0.50
Soldier Lake (SL)	28	37	0.64	0.67
Bay de Wasai (BDW)	21	28	0.36	0.53
Churchville Point (CHP)	27	35	0.58	0.60
Whipple Point (WP)	28	34	0.57	0.58
Cook Island (CI)	7	23	0.49	0.63
Cedar Point (CP)	8	26	0.54	0.61
Mission Creek (MC)	11	24	0.46	0.53

Table 5
Pairwise Fst values for yellow perch: 11-Mile Marsh (EMM), Palmer's Point (PP), Ashmun Bay (AB), Soldier Lake (SL), Bai de Wasai (BDW), Churchville Point (CHP), Whipple Point (WP), Cook Island (CI), Cedar Point (CP), and Mission Creek (MC).

Population	EMM	PP	AB	SL	BDW	CHP	WP	CI	CP
PP	0.0005								
AB	0.0470	0.0208							
SL	0.1918*	0.2224*	0.2903*						
BDW	0.0107	0.0140	0.0258	0.2492*					
CHP	0.1237*	0.1540*	0.2188*	0.2035*	0.1814*				
WP	0.0930*	0.1477*	0.2314*	0.2250*	0.1688*	0.0547			
CI	−0.0248	0.0033	0.0265	0.2207*	−0.0272	0.1575*	0.1189*		
CP	−0.0148	0.0270	0.0929	0.2000*	0.0157	0.0772*	0.0572	−0.0215	
MC	0.0295	−0.0107	0.0289	0.2505*	0.0338	0.1614*	0.1586*	0.0403	0.0421

* Significant ($P<0.05$) following sequential Bonferroni correction for multiple tests.

(2006) found that because of their use of specific spawning shoals, native walleye in the New River, VA persisted despite significant stocking from outside, lake-derived sources.

Yellow perch

The genetic diversity of yellow perch was generally lower than that for walleye (Stepien and Faber, 1998; Stepien et al., 2004). Because of this low diversity and high frequencies of straying during spawning, fine-scale stock structure does not commonly develop in yellow perch (Miller, 2003; Glover et al., 2008; Sepulveda-Villet et al., 2009). In the SMR, the mean heterozygosity observed for each yellow perch population across all loci was lower than expected values. Although they used different loci, a similar trend was observed by Parker et al. (2009) for yellow perch samples collected in lakes Huron and Michigan. Reduced heterozygosities may be the result of inbreeding and small population sizes, or the Wahlund effect due to mixing among populations (Hallerman, 2003). Inbredding does not appear to be the case in the SMR. The size of the perch population in the SMR as a whole has been largely stable since the 1970s. It experienced rapid growth in the late 1980s, but there have been no documented declines that would lead us to believe that inbreeding was the cause for the low heterozygosities (Schaeffer et al., 2011). The low heterozygosities observed may have been caused by low sample sizes at some locations, or the sampling of non-spawning fish may have led to mixing of populations (Gharrett and Zhivotovsky, 2003). The low heterozygosities also may have been a function of the loci used in this study; the population of yellow perch in Saint-Francois Lake, Quebec, that was used to develop the Pfla primers also exhibited lower heterozygosities than expected (Leclerc et al., 2000).

The yellow perch population with the highest heterozygosity in our study was that from Soldier Lake. This small (7.6 ha) inland lake is isolated from the SMR, and we expected to observe genetic differences between this population and the SMR populations. However, we did not expect that the diversity in Soldier Lake would exceed that of the SMR. Despite the small size of the lake, it has an abundant population that has been targeted for manual removals of yellow perch to improve size structure. Even after such removals, the population was still estimated to be 79,000 adult individuals (Bauer et al., 2004).

Two yellow perch populations within the SMR system, Churchville Point and Whipple Point, were genetically distinct from those in the rest of the system. These populations are both located in Lake George, the same location as the unique Bar River walleye population. Lake George is a lentic area of the SMR, a habitat distinct from the other sampling points. Although fine-scale structure may not be expected in yellow perch populations (Ford and Stepien, 2004), Parker et al. (2009) observed genetic differences between perch that occupied different habitats in southern Lake Michigan. Spawning perch were not targeted in either our study or by Parker et al. (2009), which may have resulted in increased mixing of populations. However, further study targeting spawning individuals and collecting larger sample sizes per group and more loci screened may allow clearer patterns in the population structure to be recognized.

Lake George isolation

Both walleye and yellow perch had unique populations in Lake George. Similar results also have been found with a population of lake sturgeon (*Acipenser fulvescens*) from the same site. This lake sturgeon population appears to be distinct from populations in the rest of the Great Lakes (J. Comben, Lake Superior State University, unpublished data). It is unclear why this area of the river appears genetically distinct from the rest of the system. Currently, there are no apparent physical barriers preventing movement of fish in and out of this area. However, these populations may reflect a relatively recent isolation, as this area may have been separate from the main SMR prior to geological rebound. Geologically, the SMR is a young system. The St. Marys Rapids were created only 2000 years ago, when the elevations of Lakes Huron and Michigan dropped. The history of these drainage patterns may have significant implications for the population genetics of fish species native to this area, and further investigation is warranted. During the last ice age, walleye populations in Lake Erie were thought to be separated by ice sheets (Strange and Stepien, 2007). Once these ice barriers broke down, genetic differences were maintained by natal site fidelity. Possibly, a similar mechanism is at work in the Lake George region of the SMR.

Stocking effects

Because historical samples from the SMR were not available, we cannot directly measure the full extent of genetic change due to stocking. The unique genetic signatures of the Bar and Charlotte River populations suggest that there may have been other unique walleye populations in the SMR before stocking began, and further sampling of smaller spawning populations throughout the river may reveal that more unique populations still remain. Stocking is one of the most important and visible tools for managing fish populations (Schramm and Piper, 1995), and often hatcheries are called upon to restore depleted stocks or enhance existing fisheries through supplementation (Miller and Kapuscinski, 2003). Historically, this activity neglected the potential genetic impacts on native fish populations, resulting in genotypic change of some native populations. However, in many current stocking practices, the genetic composition of both the source and recipient populations is considered. This ensures that stocking will have a better chance of success and that unique populations are not lost. This latter type of stocking strategy is now employed in the SMR, as only local broodstocks are used for supplementation. However, possible local adaptations of unique stocks, such as those in Bar River, are not considered.

This study represents the first attempt to characterize the population genetic structure of walleye and yellow perch populations in the SMR. It provides a baseline upon which more detailed research can build. By focusing on collecting larger numbers of spawning fish from more populations and analyzing variation at more loci, future work could

provide more information about population genetic differentiation in this diverse system. While stocking has not been a strategy employed for managing yellow perch populations, it has influenced the genetic diversity of the walleye populations. Although the Munuscong Bay population is used as the current walleye broodstock, the potential for genetic change is still present given the unique populations that exist in the SMR. The uniqueness of the fish populations in Lake George suggests that populations there should be protected from future stocking events, and further work is encouraged to understand the nature of its genetic isolation, despite its physical connectedness.

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